Kyle Zeman

Buffalo, NY | U.S. Citizen | (585) 280-3285 | kylezeman05@gmail.com | linkedin.com/in/kyle-zeman | kylezeman.vercel.app

Mechanical engineering student with quality and process internship experience in precision manufacturing and strong SolidWorks modeling, drawing and FEA skills. Interested in design, analysis and manufacturing work on aerospace and defense hardware. U.S. citizen, able to obtain a security clearance.

Education

University at Buffalo, The State University of New York — Buffalo, NY
Bachelor of Science in Mechanical Engineering — *Expected May 2027*
Relevant coursework: Manufacturing Processes, Design Process & Methods, Machines & Mechanisms, Additive Manufacturing, Dynamic Systems, Fluid Mechanics & Lab, Numerical Methods
Peer tutor: Mechanics of Materials and SolidWorks design courses

Experience

Handshake AI Fellowship | AI Evaluation & Training Fellow — *2026 to Present*
Remote

- Check AI-generated engineering and technical content for accuracy against set criteria.
- Write prompts and scoring rubrics for large language models and give annotated feedback on responses.

Vishay Thin Film | Quality / Process Engineering Intern — *June 2026 to Aug 2026*
Niagara Falls, NY

- Led Layered Process Audits (LPA) across 8 manufacturing processes with little guidance, the first step of a plant-wide effort to find noncompliances, and drove corrective actions that improved efficiency and reduced defects.
- Ran meetings with operators and engineers to plan how each improvement would be put in place and to stop recurring nonconformances.

PayQuicker | Tax Documentation Specialist — *July 2023 to May 2026*
Remote

- Reviewed and approved high volumes of tax documentation for accuracy and regulatory compliance.
- Built Microsoft Power BI reports to track documentation status and processing trends.

Engineering Projects

Adjustable-Deadrise Keel | Senior Design Capstone — *Fall 2026 to Spring 2027*
MAE 451 / 494, Industry Sponsored

- Work on a six-person team designing a flexible keel that adjusts deadrise from 10° to 35° for an 8 m military / Coast Guard shore-approach craft.
- Plan composite and hinge bend tests and build FEA models to predict keel stress and deflection.

Corvette C5 Aerodynamic CFD Analysis | SolidWorks Flow Simulation — *Summer 2026 to Present*

- Modeled the full exterior body of a Corvette C5 using surfacing to capture the curved geometry.
- Setting up external-flow CFD simulations to view airflow around the vehicle and evaluate aerodynamic performance (in progress).

Scissor Car Jack Reverse Engineering & FEA | SolidWorks — *Fall 2025*

- Measured and modeled each component of a scissor jack into a fully constrained assembly.
- Ran finite element stress simulations under rated load to find the factor of safety.

Nerf Blaster Reverse Engineering | Mechanical Engineering Project — *Fall 2024*

- Worked with a team to take apart and analyze the components of a Nerf blaster.
- Proposed ways to manufacture it more efficiently and improve product quality.

Technical Skills

CAD: SolidWorks (part and assembly modeling, surfacing, GD&T, drawings), Autodesk Fusion 360
Analysis: Finite Element Analysis (SolidWorks Simulation), CFD (SolidWorks Flow Simulation), Hydraulic Calculations, Reverse Engineering
Quality & Manufacturing: Layered Process Audits (LPA), Corrective Action, Nonconformance Reduction
Fabrication & Inspection: 3D Printing, Machine Shop Tools, Measurement and Inspection Tools
Programming & Data: MATLAB, Python, Arduino IDE, Excel, Microsoft Power BI

Certifications: Certified SolidWorks Associate (CSWA), in progress